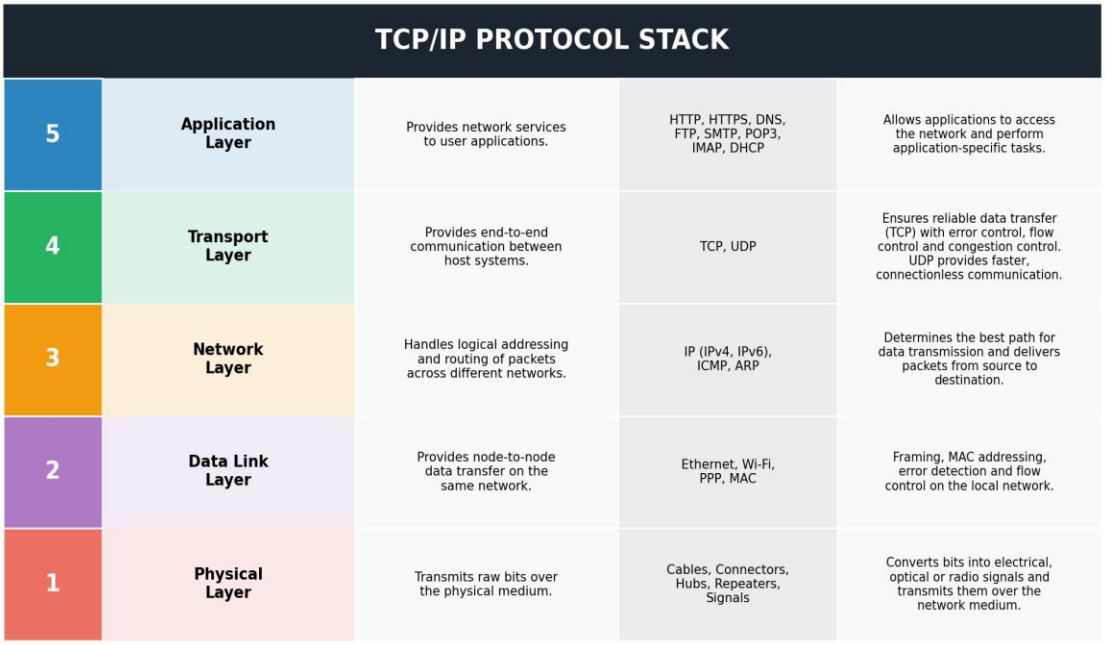
ASSIGNMENT – 1

PCCST501 – Computer Networks

Title: How the Internet Delivers a Web Page

Website Chosen: https://www.instagram.com

1. Draw the TCP/IP protocol stack and explain the role of each layer.



Explanation of the TCP/IP Protocol Stack Diagram

The TCP/IP Protocol Stack is a communication model that defines how data is transmitted over the Internet. It consists of five layers, and each layer performs a specific function. Together, these layers ensure that data is transmitted efficiently, accurately, and securely from one device to another.

1. Application Layer

The Application Layer provides network services directly to user applications. It enables applications such as web browsers, email clients, and file transfer programs to communicate over the Internet. Common protocols used in this layer include HTTP, HTTPS, DNS, FTP, SMTP, POP3, IMAP, and DHCP. When a user enters www.instagram.com in a browser, this layer generates the HTTP/HTTPS request.

2. Transport Layer

The Transport Layer provides end-to-end communication between devices. It uses TCP (Transmission Control Protocol) for reliable communication and UDP (User Datagram Protocol) for faster communication when speed is more important than reliability. TCP ensures that data such as images and videos on Instagram is delivered correctly, in the proper order, and without loss.

3. Network Layer

The Network Layer handles logical addressing and routing of data packets across different networks. It uses protocols such as IP (IPv4 and IPv6), ICMP, and ARP. This layer determines the best path for transmitting packets from the source to the destination and ensures they reach the correct device.

4. Data Link Layer

The Data Link Layer provides node-to-node communication within the same local network. It is responsible for framing, MAC addressing, and error detection. Technologies such as Ethernet, Wi-Fi, PPP, and MAC protocols operate at this layer to ensure reliable communication between directly connected devices.

5. Physical Layer

The Physical Layer is the lowest layer of the TCP/IP stack. It transmits raw bits over the physical communication medium using cables, optical fiber, wireless signals, connectors, hubs, and repeaters. It converts digital data into electrical, optical, or radio signals for transmission.

2. Explain the role of DNS in converting the domain name into an IP address.

DNS (Domain Name System) translates a human-readable domain name into an IP address that computers can understand.

Example:

```
www.instagram.com
      |
      ▼
DNS Resolver
      |
      ▼
DNS Server
      |
      ▼
157.240.xxx.xxx
```

Working of DNS

- User enters www.instagram.com in the browser.
- Browser checks its DNS cache.
- Operating System checks local cache.
- Request is forwarded to the DNS Resolver.
- DNS Server returns the IP address.
- Browser establishes a connection using the received IP address.

Importance of DNS

- Converts domain names into IP addresses.
- Makes Internet browsing easier.
- Eliminates the need to remember numerical IP addresses.

3. Describe how HTTP is used to request and receive the webpage.

After receiving the IP address, the browser communicates with the web server using HTTP or HTTPS.

Steps

- Browser establishes a TCP connection.
- HTTPS encrypts communication using TLS.
- Browser sends an HTTP GET request.

Example:

```
GET / HTTP/1.1
Host: www.instagram.com
```

- Web server processes the request.

Server sends back:

- HTML
- CSS
- JavaScript
- Images
- Videos
- Browser renders the webpage (Instagram feed) for the user.

4. Explain how TCP provides reliable communication for HTTP.

TCP (Transmission Control Protocol) ensures reliable delivery of data.

TCP Three-Way Handshake

```
Client                          Server
SYN ---------------------->
     <---------------- SYN + ACK
ACK ---------------------->
      Connection Established
```

Features of TCP

- Reliable communication
- Error detection
- Packet retransmission
- Ordered delivery
- Flow control
- Congestion control

These features ensure that the Instagram webpage and media content load correctly without missing or corrupted data.

5. Identify whether the application follows a Client–Server or Peer-to-Peer architecture and justify your answer.

Architecture Used: Client–Server

```
Browser (Client)
       |
  HTTP Request
       |
       ▼
Instagram Web Server
       |
  HTTP Response
       ▼
Browser Displays Webpage
```

Justification

- Browser acts as the client.
- Instagram acts as the server.
- Client requests webpages, images, and videos.
- Server processes the request and sends the response.

Therefore, Instagram follows the Client–Server Architecture.

6. Compare HTTP and FTP with at least four differences.

HTTP	FTP
Used for accessing web pages	Used for transferring files
Default Port: 80 (HTTP), 443 (HTTPS)	Default Port: 21
Stateless protocol	Session-oriented protocol
Mainly transfers HTML, CSS, images	Transfers files and folders
Used by web browsers	Used by FTP clients

7. Explain how cookies improve user experience with one real-world example.

Cookies are small text files stored by a web browser on a user's device when they visit a website. They help websites remember information about the user, making future visits faster, more convenient, and personalized.

How Cookies Improve User Experience

- Remember login information, so users do not need to sign in every time.
- Store user preferences, such as language and theme settings.
- Keep track of recently viewed content.
- Provide personalized content and recommendations based on browsing history.

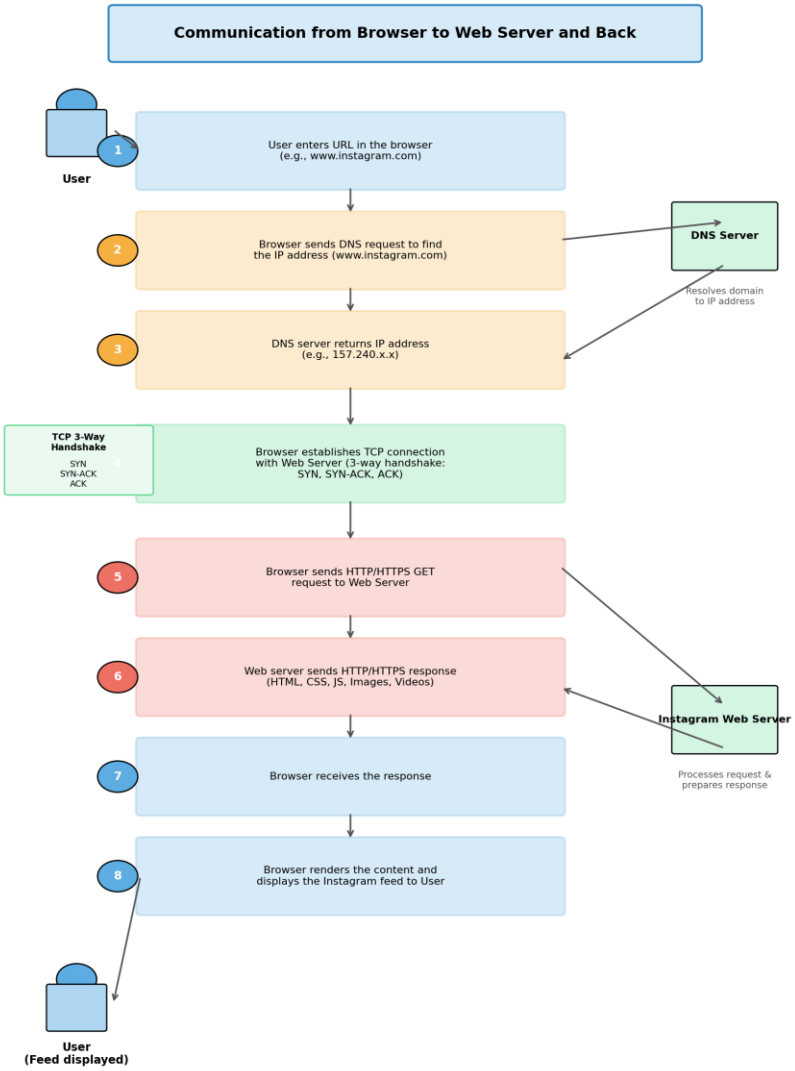
Real-World Example (Instagram)

When a user logs in to Instagram, the website stores a cookie in the browser. The next time the user visits Instagram, the cookie allows the website to:

- Keep the user logged in without entering the username and password again.
- Display a personalized feed and Explore page based on past activity.
- Remember followed accounts and recent searches.
- Save preferences such as dark mode and notification settings.

As a result, the user enjoys a faster, more personalized, and seamless browsing experience.

8. Draw a flow diagram showing the complete communication from the browser to the web server and back.



9. Write a brief conclusion (100–150 words) describing the importance of application-layer protocols in everyday Internet usage.

Conclusion

Application layer protocols play a vital role in everyday Internet communication by enabling users to access websites, send emails, transfer files, and use online services efficiently. Protocols such as HTTP/HTTPS, DNS, FTP, and SMTP ensure smooth and secure communication between users and servers. In the case of Instagram, DNS converts the domain name into an IP address, HTTP/HTTPS securely delivers the webpage along with images and videos, and cookies improve the browsing experience by remembering user preferences and login details. These protocols work together with the TCP/IP model to provide reliable, fast, and secure data transmission. Therefore, application layer protocols are essential for making Internet services simple, efficient, and user-friendly.